

Thesis/Project and Intern Management System

by

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*Project (CSE 400) submitted in partial fulfillment of the
requirements for the degree of*

Bachelor of Science in Computer Science and Engineering

Under the supervision of

Md.Rajibul Palas



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DECLARATION

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We declare that this project entitled *Thesis/Project and Intern Management System* is the result of our own work except as cited in the references. The project has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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Based on the evaluation and discussion held on **September 26, 2025**, we, the undersigned, hereby endorse the project for acceptance and approval.

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MD Abid Hasan, Full Kumar Roy and Manik Chandra Roy

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Dedicated to
my loving Mother
Mst. Anjumanara Begum

– MD Abid Hasan

Dedicated To my beloved
Loving Parents
Gobindo Roy, Bishoka Rani

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Dedicated To my beloved
Loving Fathers
Suzon Chandra Roy

– Manik Chandra Roy

ABSTRACT

In most universities, the process of managing students' thesis projects and internship programs is complex and often handled manually, leading to inefficiencies, miscommunications, and delays. This project titled **"Thesis/Project and Intern Management System"** is designed to digitize and simplify the entire process of thesis and internship management for students, faculty members, and administrators. The system provides an integrated platform where students can easily submit their proposals by selecting the department, academic year, semester, course type, project title, problem statement, and group members. For internship students, the system enables them to provide their own company details or select from a list of university-approved companies with available slots. The proposed approach follows a role-based workflow. Students begin the process by submitting proposals, which are first reviewed by the administrator and then sent to the Research Cell for evaluation. After being approved by the Research Cell, the proposals are assigned to supervisors who guide the students throughout the thesis or internship journey. The administrator has full control over the system management, which includes creating sections, allocating group sizes, creating research cells, and approving proposals. A clear division of responsibilities is maintained so that a faculty member assigned to a research cell cannot act as a supervisor for the same project, ensuring fairness and transparency. The expected outcomes of this system include improved efficiency in proposal management, reduced paperwork, and improved communication between students, supervisors, and the research cell. The system also ensures transparency in the approval process and allows for real-time status tracking for students. Ultimately, the goal of this system is to create a structured, reliable, and scalable solution that improves the overall quality of academic project and internship management within the university.

Keywords: project management, thesis management, intern management, academic system

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1. Introduction

1.1 Background

Managing thesis, project, and internship programs in universities is often a complex and manual process. Students submit proposals, supervisors guide them, and administrators manage approvals and documentation. This manual system can lead to inefficiencies, miscommunication, and delays in project execution[1]. Digitalization of this process provides a streamlined and transparent platform for all stakeholders, ensuring better organization, real-time tracking, and improved communication [2].

1.2 Problem Statement

The current system lacks an integrated approach to manage thesis, project, and internship submissions. Students face difficulties in submitting proposals, supervisors have limited tracking of progress, and administrators struggle to maintain records efficiently[3]. The main problem addressed by this project is to design a system that allows:

- Students to submit proposals and track their status.
- Research Cell and Supervisors to review, approve, and guide students effectively.
- Administrators to manage departments, research cells, groups, and proposals efficiently.

1.3 Purpose of the Project

The objective of this project is to develop a **Thesis/Project and Intern Management System**. that:

- Provides a role-based workflow for students, faculty, and administrators [4].
- Ensures transparency in proposal approval and project tracking.
- Minimizes paperwork and manual interventions.
- Facilitates internship management with both student-suggested and university-assigned companies.
- Generates real-time status updates for students and faculty members.

1.4 Historical Context

Previously, thesis and internship management were handled manually with paper-based submissions. Supervisors had limited visibility of student progress, and administrators struggled to maintain proposal lists and group assignments. This led to inefficiencies, lost documentation, and delays in project completion. Various universities have now started adopting digital management systems to improve workflow efficiency and communication between stakeholders.

1.5 System Overview

The system follows a role-based workflow:

- **Students** submit proposals with all required information, including department, academic year, course details, group members, project title, problem statement, and motivation.
- **Administrators** review submissions, create departments, assign research cells, set group sizes, and forward proposals to research cells.
- **Research Cell** evaluates proposals, approves them for supervisors, or sends them back for corrections.

- **Supervisors** guide approved students, track progress, and provide feedback.
- For internships, students can either suggest their own companies or select from university-approved companies, including industrial supervisor details [5].

Table 1.1: Roles and Responsibilities in the System

Role	Responsibilities
Student	Submit proposals, track status, select skills/company
Admin	Manage departments, groups, research cells, approve proposals
Research Cell	Evaluate proposals, forward for supervision, send back corrections
Supervisor	Guide students, monitor progress, approve/reject final submission

2. Literature Review

A literature review provides a comprehensive synthesis of existing scholarly works, including research articles, books, and other relevant sources, related to thesis, project, and internship management systems in academic institutions. It establishes the context for the proposed Thesis/Project and Intern Management System (TPIMS) by identifying current challenges, gaps, and trends in existing systems.

2.1 Related Work

Previous research has shown that managing the student thesis and project submission process manually or through spreadsheets often leads to inefficiencies, lost documents, and delayed approvals. Some universities have implemented digital platforms for submission and tracking; however, there is often a lack of integration between students, supervisors, research cells, and administrators. Role-based access and real-time tracking are limited in many existing systems [3, 4]. Internship management has traditionally been separate from thesis/project workflows, with manual registration, limited real-time updates, and difficulty monitoring available slots [6, 7]. Role-based workflows in academic systems are essential to ensure individual responsibility, reduce conflict, and maintain transparency, with research cell evaluation, supervisor guidance, and administrator approval being important components [8, 2]. Digital internship management platforms facilitate communication between students, institutions and companies, improve skills development and managerial efficiency [6]. Role-based access control provides secure and efficient data handling, while modern web-based systems increase overall efficiency, reduce administrative overhead and allow for better tracking of student progress [9, 1, 10]. These findings highlight the need for integrated, role-based, real-time academic management platforms.

Table 2.1: TPIMS vs Existing Academic Systems

System	Core Features	Advantages	Limitations
Basic Spreadsheet System	Proposal submission, manual tracking	Easy to set up, low cost	Prone to errors, slow approval process, no integration
University Portal	Submission & approvals	Some integration	Limited role-based access, no internship management
TPIMS (Proposed)	Thesis/project + internship management, role-based, automated workflow, real-time updates	Integrated, transparent, reduces manual work, efficient	Requires initial setup, training for users

2.2 Research Gap & Our Contribution

While previous solutions address the separate components of thesis, project or internship management, they lack a unified, integrated system that:

- Integrates thesis/project and internship management on a single platform.
- Supports role-based access for students, supervisors, research cells and administrators.
- Provides real-time status updates and automated workflow routing.
- Allows structured management of internship programs by companies and industry supervisors.

To bridge this gap, the proposed TPIMS integrates thesis, project and internship management into a single system. It provides role-based workflows, real-time tracking, automated notifications and structured internship placement, increasing efficiency, transparency and communication in academic project management [11, 12].

2.3 Limitations of Existing Systems

Existing systems in educational institutions face several recurring limitations:

- Lack of integration between thesis/project and internship management.
- Limited role-based access, creating conflicts of interest or security issues.
- Manual submission and tracking, leading to delays and lost documentation.
- Insufficient reporting and analytics for administrators and supervisors.
- Minimal support for managing student-selected internships or companies.

Table 2.2: Traditional Systems vs. TPIMS Approach

Component	Traditional System	TPIMS Approach
Proposal Submission	Manual forms, email, or spreadsheet	Digital submission via system with real-time tracking
Supervisor Assignment	Random or manual assignment	Role-based assignment with conflict prevention
Internship Management	Separate manual process	Integrated within system, includes student-suggested or university-approved companies
Reporting	Manual or limited	Automated reports and analytics for students, supervisors, and admins
Access Control	Same access for all users	Role-based access for students, supervisors, research cells, and admins

TPIMS addresses these limitations by offering an integrated, digital, and role-based platform that combines thesis/project and internship management to provide real-time status updates, workflow automation, and structured internship placement [13, 3, 4].

3. Methodology

The methodology outlines the systematic approach used to design, develop, and implement the **Thesis/Project and Intern Management System**. This section describes the research design, system development techniques, data collection, and analysis methods applied in the project. The main goal is to create a reliable, efficient, and role-based system for managing student thesis, project, and internship workflows [14].

3.1 Project Planning

We adopted an Agile methodology for this project, focusing on flexibility and iterative development. The process was divided into two-week sprints, which allowed for regular updates and feedback. The timeline was visualized using a Gantt chart, which mapped out each phase of the project from inception to deployment. This approach allowed the team to maintain clarity about deadlines, responsibilities, and task dependencies, ensuring that the project progressed smoothly without confusion.

3.1.1 SDLC Model Comparison

We evaluated various software development life cycle (SDLC) models to find the one that best suited the project needs. The Agile model was selected because it allowed for continuous feedback and frequent updates, which were essential for the evolving nature of TPIMS. Here is a comparison of the various SDLC models in terms of flexibility, cost efficiency, risk management, and ease of implementation:

SDLC Model	Best For	Advantages	Disadvantages
Agile	Projects needing regular updates, feedback	Highly flexible, Frequent deliveries, Close team & client interaction	Needs active collaboration, Not ideal for large teams
Waterfall	Simple or well-defined projects	Clear structure, Easy to manage for beginners	Hard to change once started, Delayed testing
V-Model	Safety-critical or test-driven applications	Strong focus on verification and validation	Rigid structure, No early prototyping
Spiral	High-risk or research-heavy projects	Excellent risk handling, Emphasizes validation at each step	Complex to manage, Costly for small teams
Iterative	Medium-complexity apps with gradual refinement	Delivers working versions early, Encourages feedback-driven design	May lead to rework if base version isn't strong

Table 3.1: Comparison of SDLC Models

3.1.2 SDLC Model Score Comparison

Given the Agile model's high score in terms of flexibility and ease of implementation, it was selected as the most suitable for this project.

SDLC Model	Flexibility	Cost Efficiency	Risk Management	Ease of Implementation
Agile	5	3	4	4
Waterfall	2	5	3	4
V-Model	1	3	5	3
Spiral	3	2	4	2
Iterative	4	3	3	5

Table 3.2: SDLC Model Score Comparison

3.2 Project Timeline

Data required for the system includes: The development process was conducted following a structured timeline, which was clearly outlined in a Gantt chart. The project phases were carefully planned so that tasks were completed on time and the project progressed smoothly from one phase to the next.

3.2.1 Gantt Chart

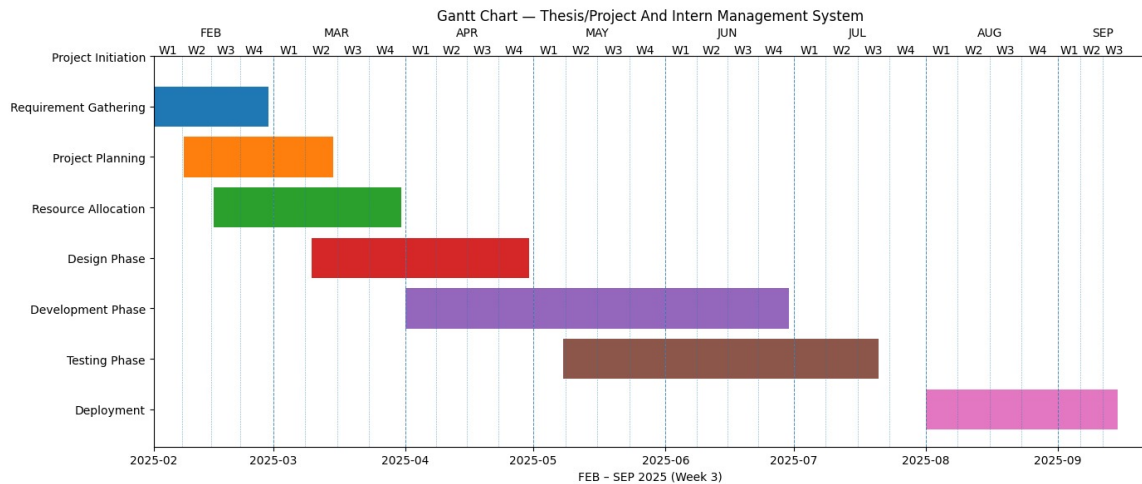


Figure 3.1: Gantt Chart showing the timeline and phases of the project.

The project timeline from February 2025 to September 2025 is as follows:

1. **Project Initiation (February 2025):** The team was formed, the project goals were set, and initial planning was done [15].
2. **Requirements Gathering (February - March 2025):** We reached out to students, faculty, and administrators to gather requirements and understand the needs of the system.
3. **Project Planning (March 2025):** A detailed project plan was created, outlining tasks, timelines, and responsibilities for the team.
4. **Resource Allocation (March 2025):** Tools, technologies, and resources were selected, and work was divided among team members.
5. **Design Phase (April - May 2025):** We designed the system's structure, database schema, and user interface. Prototypes and mockups were created for review.
6. **Development Phase (May - August 2025):** The system was developed in phases, with each sprint focusing on different features such as proposal submission, internship placement, and role management.

7. **Testing Phase (August - September 2025):** The system has undergone rigorous testing, bugs have been identified, and all features have been confirmed to be working properly in various roles.
8. **Deployment (September 2025):** Final preparations for deployment, which include writing documentation, preparing for presentation, and deploying the system.

3.3 Ethical Considerations and Limitations

The system ensures:

- Privacy and security of student and faculty data using Laravel authentication and role-based access control [16].
- Conflict of interest prevention by restricting Research Cell members from supervising the same project.
- Limitation: The system depends on accurate data entry and may require manual intervention in exceptional cases.

4. System Requirements

4.1 Functional Requirements

To ensure that TPIMS effectively meets the needs of students, faculty members, and administrators, several essential features are outlined:

- **Role Management and User Login:**

- The system is designed to allow staff to log in based on their role (administrator, faculty, supervisor, student, etc.).
- Role-based access ensures that each user will have access to only the relevant features and data.

- **Proposal Submission and Review:**

- Students can submit their thesis proposal, which includes details such as department, academic year, semester, project title, problem statement, and group members.
- For internship placements, students can select from a list of university-approved companies or provide their own company details.
- Administrators review the proposals and forward them to the Research Cell for approval [2].

- **Assignment of Supervisors:**

- After proposals are approved by the research room, administrators assign supervisors to guide students through the thesis or internship process.
- This system ensures that no faculty member can serve as a supervisor for projects within their assigned research room, ensuring fairness and transparency.

- **Real-time Status Tracking:**
 - Students and faculty members can track the status of proposals and projects in real-time.
 - The system provides notifications for each stage of the proposal submission, review, and approval process.
- **Reporting and Analytics:**
 - Administrators can generate reports on the status of proposals, track student progress, and view analytics related to project completion.
 - The system can generate summary reports, which allow faculty to monitor the overall progress of thesis and internship projects.

4.2 Non-functional Requirements

In addition to functional features, TPIMS must meet several non-functional requirements to ensure its reliability, scalability, and usability:

- **Fast operation:**
 - The system must operate quickly to ensure that users can submit proposals, track progress, and manage data without delay.
- **Security:**
 - Role-based access control ensures that only authorized users can access sensitive data.
 - Secure data storage and encryption protocols are used to protect user data and ensure confidentiality.
- **Usability:**
 - The system is designed to be user-friendly, with an intuitive interface that reduces training time for staff and students.
 - A responsive UI ensures that the system can be accessed from both desktop and mobile devices.

- **Offline functionality:**
 - The system allows users to work offline if needed, with data synchronization occurring once the internet connection is restored.
- **Maintainability:**
 - The system is built with a modular architecture, allowing for easy updates and modifications without affecting overall performance.
- **Scalability:**
 - The system is designed to adapt to the needs of the university, allowing for the addition of additional users, departments, or campuses without significant changes to the core architecture [11].

4.3 System Architecture

TPIMS is based on a client-server architecture, with a frontend built using **Tailwind CSS** to create a dynamic, responsive user interface and a backend powered by the **PHP Laravel** framework for data management and real-time interaction.

- **Frontend:** Tailwind CSS provides a flexible and customizable UI framework that creates a clean, responsive design that works well across a variety of devices. It enables rapid styling with utility-first classes, ensuring the interface is intuitive and user-friendly[17].
- **Backend:** The backend is built using the PHP Laravel framework, a powerful, MVC-based framework that simplifies routing, data management, and security. Laravel's built-in features such as authentication, routing, and session management help streamline development and ensure maintainable code.
- **Database:** MySQL is used as a relational database management system (RDBMS) to store all data related to proposals, users, supervisors, and project status. MySQL provides high performance, reliability, and ease of use for relational data [12].

- **API:** The system integrates with external tools and services (such as email notifications) via RESTful APIs, which ensures smooth communication between the frontend, backend, and external systems [10].

4.4 Tools and Technologies

The following tools and technologies are used to develop and deploy TPIMS:

- **Backend:** PHP Laravel framework to handle backend logic and support data management [18].
- **Frontend:** Tailwind CSS to create a responsive and visually appealing user interface.
- **Database:** MySQL to manage data related to proposals, users, projects, and status.
- **Version Control:** Git and GitHub for version control and collaborative development [19].
- **Development Environment:** Visual Studio Code as the main code editor and a localhost server for local development and testing [20].

4.5 Budget Estimate

The following budget breakdown provides an estimate for the development of TPIMS:

Budget Category	Estimated Amount (BDT)
Human Resources	200,000
Software & Tools	6,000 – 8,000
Hardware	15,000
Miscellaneous & Support	10,000
Total Estimated Cost	231,000 – 233,000

Table 4.1: Budget Breakdown for TPIMS

4.5.1 Cost Estimation

4.5.1.1 Human Resource Expenses

From Table 4.1, the team includes three core developers working over six months. Costs cover development, testing, and documentation:

- Developers: $3 \times \text{BDT } 10,000/\text{month} \times 6 \text{ months} = \text{BDT } 180,000$
- UI/UX Design: Interface design and user flow development = **BDT 15,000**
- Project Management & Documentation: Planning and reporting tasks = **BDT 5,000**

Subtotal (Human Resources): BDT 200,000

4.5.1.2 Software and Tools

From Table 4.1, most tools are open-source, with minor expenses for deployment:

- Domain (.com): For hosting/presentation = **BDT 2,000**
- Web Hosting (optional): Shared/VPS = **BDT 4,000 – 6,000**

Subtotal (Software & Tools): BDT 6,000 – 8,000

4.5.1.3 Hardware Costs

Existing laptops are used. Optional expenses include (as shown in Table 4.1):

- Laptop Maintenance/Depreciation = **BDT 15,000**

Subtotal (Hardware): BDT 15,000

4.5.1.4 Miscellaneous & Support

Covers connectivity, branding, and minor contingencies from Table 4.1:

- Internet/Data Usage = **BDT 2,000**
- Marketing Materials: Brochures, mockups = **BDT 3,000**
- Contingency Buffer = **BDT 5,000**

Subtotal (Miscellaneous): BDT 10,000

5. System Design and Architecture

This chapter discusses the design and architecture of the Thesis/Project and Internship Management System (TPIMS), which was carefully planned to meet the functional needs for managing academic thesis projects and internship placements. The design process integrates features such as proposal submission, role-based access, proposal review workflow, and real-time tracking into a unified system. In this chapter, we focus on the technical framework and visual design of TPIMS, providing an overview of how each component works together to provide an efficient and seamless experience for users.

5.1 System Design

To clearly explain the system, we have used several diagrams that highlight different aspects of TPIMS. As shown in Figure 5.1, the use case diagram represents the interaction between different system roles. Figure 5.2 illustrates the sequence of activities and workflows. Figure 5.3 presents the overall flow diagram showing the process structure, while Figure 5.4 shows the ER diagram defining the database organization and relationships. Together, these diagrams provide a complete view of how TPIMS works behind the scenes.

While designing each step of the system, we ensured that it would be user-friendly, efficient, and scalable for future improvements.

- **Browser (Client Side):** Staff and students interact with TPIMS through a web browser, submit proposals using the dashboard, track progress, update project details, and generate reports.
- **Server Layer:** Handles all business logic and communication between the client-side interface and the database. Implemented using the **PHP Laravel framework** for secure and fast processing.

- **API Gateway:** Directs user requests to the appropriate backend module, ensuring scalability and modularity for future upgrades.
- **Database (MySQL):** Stores all necessary data including users, proposals, internship placements, project status, and reports. MySQL ensures reliability, performance, and structured data management.
- **Frontend (Tailwind CSS):** Built with **Tailwind CSS** for a responsive, user-friendly, and visually appealing interface across devices.
- **Backend (PHP Laravel Framework):** Handles routing, data management, authentication, and all business logic, acting as the backbone of the system.
- **Version Control (Git and GitHub):** Manages code collaboration, maintains change history, and facilitates teamwork across multiple developers.

Together, these components ensure that TPIMS runs smoothly, provides a flexible, robust, and seamless

5.2 Unified Modeling Language Diagram

The UML diagram provides a visual representation of how different users interact with the Thesis/Project and Internship Management System (TPIMS) and how its key features are interconnected. As shown in Figure 5.2, this diagram outlines the main use cases, which include submitting thesis proposals, tracking project progress.

5.2.1 Use Case Diagram

Actors (Users) Involved: As shown in the Figure 5.2

Student

- **Login:** Required to access the system.
- **Registration:** Register as a user in the system.
- **Submit Proposal:** Submit a proposal for research or project.

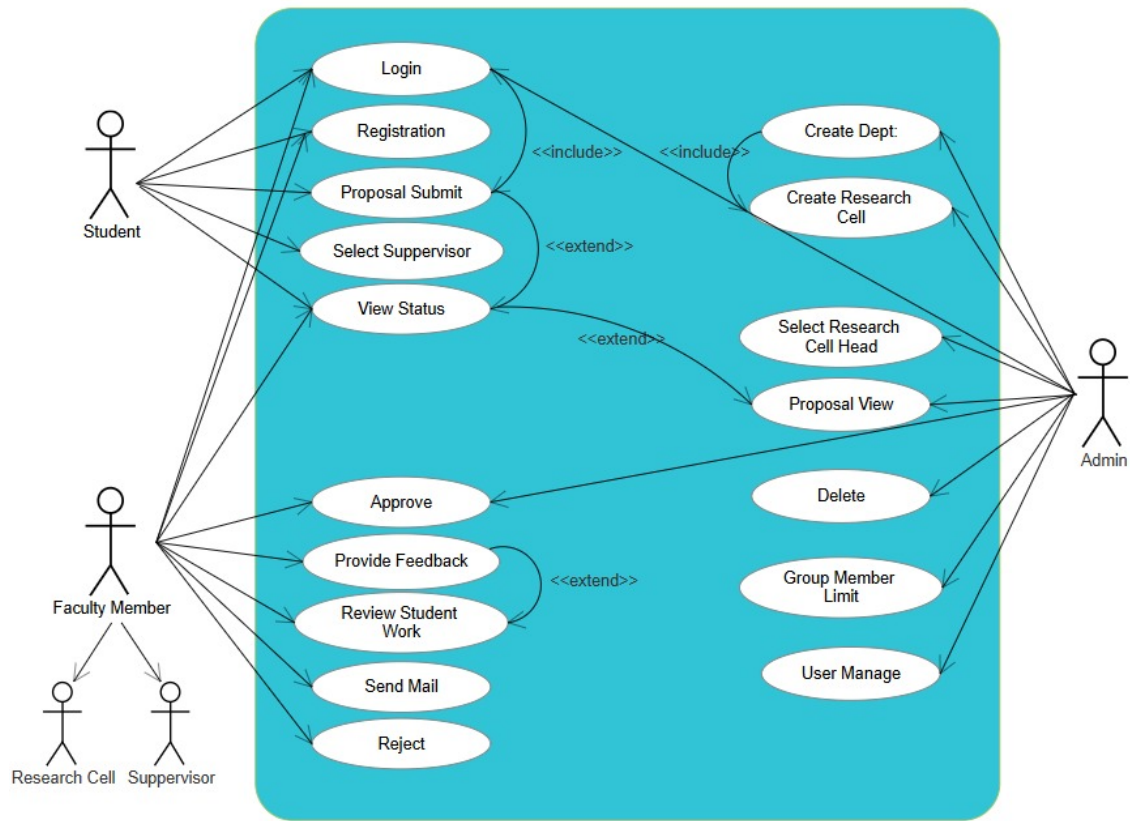


Figure 5.1: Use Case Diagram

- **Select Supervisor:** Select a supervisor for guidance.
- **View Status:** View the status of the proposal.

5.2.2 Use Case Diagram

Faculty Member

- **Approve:** Approve proposals submitted by students.
- **Provide Feedback:** Provide feedback on student work.
- **Review Student Work:** Review student submitted work.
- **Send Mail:** Send emails about student work or progress.

- **Reject:** Reject proposals or work if necessary.

Administrator

- **Create Department:** Create a new department.
- **Create Research Cell:** Establish a new research cell.
- **Select Research Cell Head:** Assign heads to research cells.
- **Proposal View:** View proposals submitted for review.
- **Delete:** Remove data or entities from the system.
- **Group Member Limit:** Set a limit for group members in the research room.
- **User Management:** Manage user roles, permissions, and profiles.

5.2.3 Relationships

- **⟨⟨include⟩⟩:** Mandatory actions that are always part of the main use case. Example: *Login* for all actors.
- **⟨⟨extend⟩⟩:** Optional actions that are triggered by specific conditions. Example: *Apply Discount* and *Tax VAT* are optional extensions of *Take Order* for the Cash Officer role.

5.2.4 Sequence Diagram

Activities and Processes

Student

- **Submit Proposal:** The student submits a project proposal.
- **Notify Update Status:** The student receives a notification about the updated status of the proposal or project.

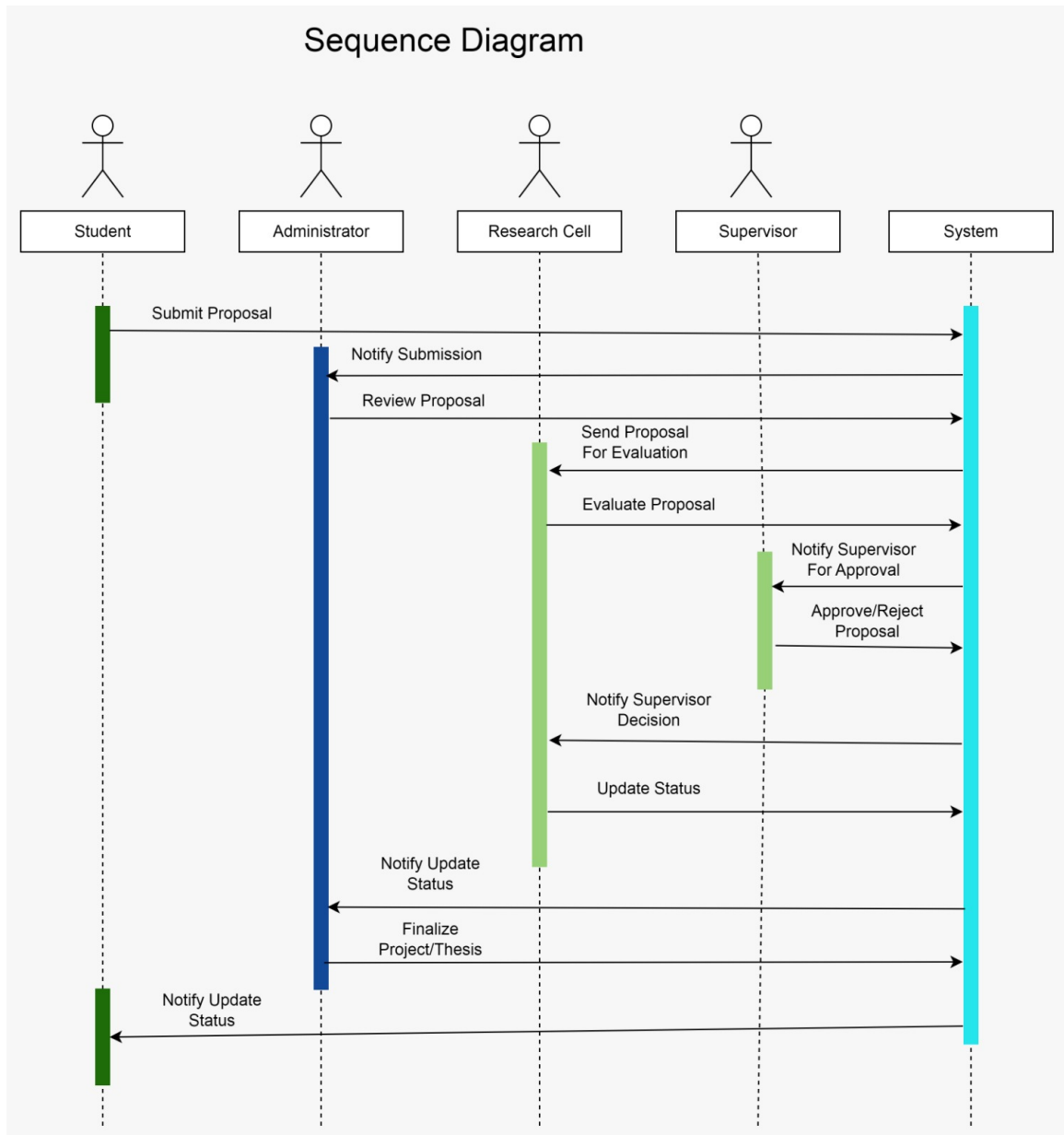


Figure 5.2: Sequence Diagram

Administrator

- **Notification of Submission:** The administrator is notified about the submission of the proposal by the student.
- **Review Proposal:** The administrator reviews the submitted proposal.

Research Cell

- **Send Proposal for Evaluation:** The research cell sends the proposal to the supervisor for evaluation.
- **Proposal Evaluation:** The research cell evaluates the proposal after receiving it.

Supervisor

- **Notify Supervisor for Approval:** The supervisor is notified to approve or reject the proposal.
- **Proposal Approval/Rejection:** The supervisor approves or rejects the submitted proposal.

System

- **Notify Supervisor for Approval:** The system notifies the supervisor about the proposal awaiting approval.
- **Notify Supervisor Decision:** The system notifies the student of the supervisor decision (whether approved or rejected).
- **Update Status:** The system updates the status of the proposal (approved or rejected).
- **Finalize Project/Thesis:** After the proposal is approved, the system finalizes the project or thesis.

5.2.5 Flow Diagram

Flow Diagram for Proposal Submission and Review Process

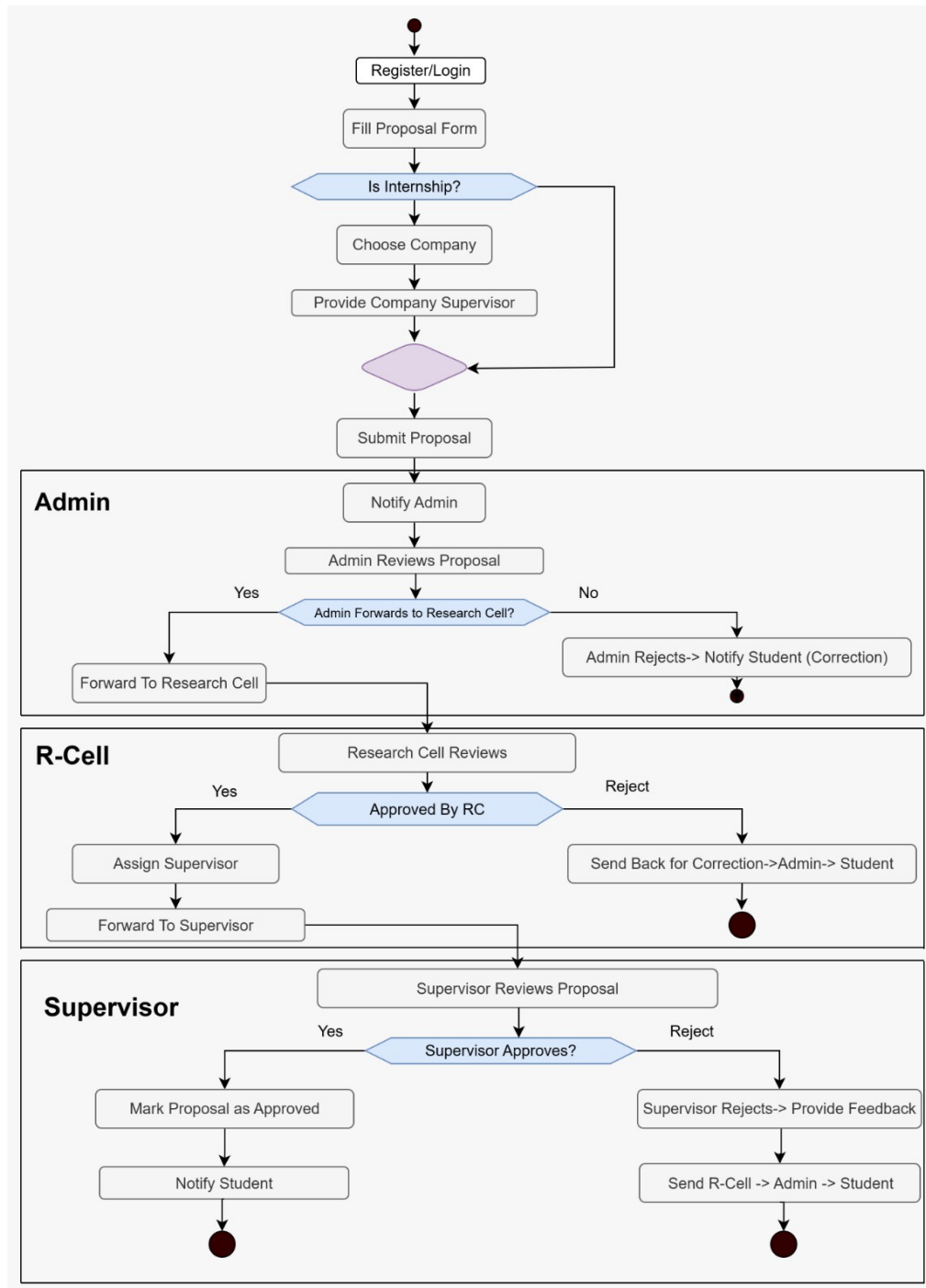


Figure 5.3: Flow Diagram

Student:

- **Register/Login:** The student registers or logs in to the system.
- **Fill in the Proposal Form:** The student fills in the proposal details.
- **What is an Internship?:** The system checks whether the proposal is related to an internship.
 - If yes:
 - * **Select Company:** The student selects a company for the internship.
 - * **Provide Company Supervisor:** The student provides information from the selected company to the supervisor.
- **Submit Proposal:** The student submits the completed proposal.

Administrator:

- **Notify Administrator:** The administrator is notified of the submitted proposal.
- **Administrator Reviews Proposal:** The administrator reviews the proposal to determine if it is acceptable.
 - If approved:
 - * **Send to Research Cell:** The administrator sends the proposal to the Research Cell for further evaluation.
 - If rejected:
 - * **Administrator Rejects:** The administrator rejects the proposal and notifies the student for revision.

Research Cell:

- **Research Cell Review:** The Research Cell reviews the forwarded proposal.
 - If approved:
 - * **Approved by RC:** The proposal is approved by the Research Cell.

- * **Forward to Supervisor:** The proposal is sent to the supervisor for approval.
- If rejected:
 - * **Send Back for Revision:** The proposal is sent back to the student for revision and the administrator is notified.

Supervisor:

- **Supervisor Reviews Proposal:** The supervisor reviews the proposal assigned to them.
 - If approved:
 - * **Supervisor Approves:** The supervisor approves the proposal.
 - * **Mark Proposal as Approved:** The proposal is marked as approved.
 - * **Notify Student:** The student is notified of the approval.
 - If rejected:
 - * **Supervisor Rejects:** The supervisor rejects the proposal.
 - * **Provide Feedback:** The supervisor provides feedback to the student on why the proposal was rejected.
 - * **Send to R-Cell, Admin, and Student:** The rejection of the proposal is communicated to the Research Cell, Admin, and Student.

5.2.6 Entity Relationship Diagram

Figure 5.4 illustrates the ER diagram of TPIMS, which defines the relational structure for managing theses, projects, research activities, and industrial proposals. The main entities include *users*, *departments*, *research cells*, *thesis/project proposals*, *industrial proposals*, *companies*, and *project members*.

- **Users** (students, faculty, administrators) have attributes such as role, email, password, department ID, and phone, and are associated with research cells and project members.
- **Departments** manage supervisors and offer thesis/project proposals when associated with research cells.

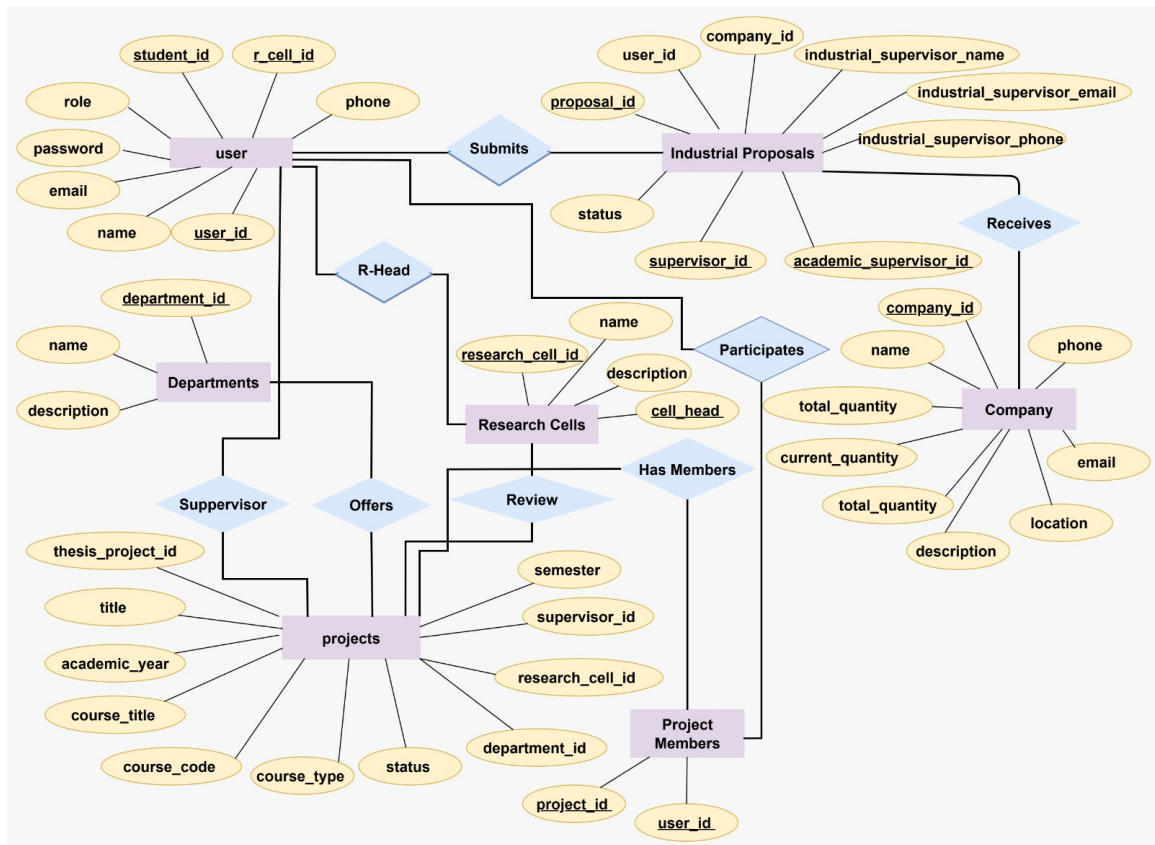


Figure 5.4: ER Diagram

- **Research Cells** have attributes such as name, description, and cell head, and they review and participate in both industrial and academic proposals.
- **Thesis/Project Proposals** include details such as title, academic year, course information, supervisor, department, and status.
- **Industry Proposals** contain internship-related details, including supervisors (academic and industry), companies, and status.
- **Companies** contain organizational information (name, contact, description, location, and supervisor details) and are associated with industry proposals.
- **Project Members** connect users to specific projects, ensuring collaboration and participation tracking.

This integrated schema supports seamless proposal submission, review, supervision, and project membership management, enabling efficient management of academic theses, research projects, and industry internships.

6. Implementation

This chapter presents the technical implementation of the Thesis/Project and Intern Management System (TPIMS). The implementation includes both backend and frontend aspects, as well as testing and deployment. The system was built using PHP Laravel for the backend, Tailwind CSS for the frontend, and MySQL as the database. GitHub was used for version control, and Visual Studio Code served as the main development environment.

6.1 Backend Implementation

The backend was implemented using the Laravel framework, which provides an MVC (Model-View-Controller) architecture, built-in security features, and efficient database management through the Eloquent ORM.

6.1.1 Data Modeling

The database schema is designed using the relational structure of MySQL. The main tables include:

- **Users** – Stores user details, authentication data, and role-based access (administrator, student, faculty, research cell).
- **Proposals** – Manages project and internship proposals submitted by students.
- **Projects** – Tracks approved thesis/project details, supervisors, and progress.
- **Internships** – Stores internship information, company details, and assigned supervisors.
- **Research Cell** – Manages faculty assignments and assigned responsibilities.

- **Groups** – Manages grouping of students for projects and internships.
- **Settings** – Stores system-wide configurations such as semester information, proposal deadlines, and notification preferences.

6.1.2 API Structure

- **Controllers** – Handle requests, responses, and perform validation.
- **Models (Eloquent ORM)** – Interact with the MySQL database.
- **Middleware** – Implement authentication, role-based permissions, and data validation.
- **Services** – Encapsulate business logic such as proposal approval, group creation, and supervisor assignment.
- **Validation** – Implemented using Laravel’s request validation system to ensure data consistency and prevent incorrect input.

6.2 Frontend Implementation

The frontend was designed using Tailwind CSS to create a responsive and modern UI. Laravel’s Blade templating engine was used to render dynamic content and integrate backend data.

6.2.1 UI and State Management

- **Role-based Dashboard** – Separate interfaces for students, supervisors, research cells, and administrators.
- **Dynamic Forms** – Implemented for proposal submission, group creation, and internship details, with validation feedback in real time.
- **Responsive Design** – Tailwind ensures a mobile-friendly layout for students and faculty.
- **Notifications** – Students receive status updates (approved, pending, rejected) through dynamic alert elements.

6.2.2 API Integration

The frontend communicates with the backend controllers via Laravel routes and AJAX/Fetch requests.

- **Students** – Submit proposals which are stored in the database.
- **Research Cell** – Reviews proposals and dynamically updates their status.
- **Supervisors** – Guide students, with updates reflected in the project tracking dashboard.
- **Administrators** – Manage all settings, groups, and approvals, with changes applied system-wide in real time.

6.3 Testing and Quality Assurance

Testing confirmed the reliability and robustness of the system:

- **Unit Testing** – Business logic such as proposal validation, role-based access, and group assignment were tested.
- **Integration Testing** – Validated database queries and approval workflows.
- **Browser Testing** – Responsiveness and usability were confirmed on Chrome, Firefox, and mobile devices.

The core workflow (proposal submission, approval chain, and supervisor assignment) was verified using automated tests in Laravel PHPUnit.

6.3.1 Environment and Deployment

Before deployment, the following steps were taken:

- Installed MySQL database and migrated schema using Laravel migrations.
- Configured `.env` files for database credentials, mail server, and security keys.
- Installed required packages via Composer (backend) and npm (frontend).
- Deployed to a cloud-based server with Apache/Nginx for production hosting.
- Managed version control via GitHub, enabling collaborative development.

7. Engineering Considerations

Engineering considerations in the design and development of the **Thesis/Project and Intern Management System** involve assessing societal impact, environmental sustainability, and ethical responsibilities to ensure a reliable and responsible solution.

7.1 Societal Impacts of Engineering Solutions

The system aims to improve communication and workflow efficiency among students, faculty members, and administrators. By providing a structured and digital platform:

- Students receive timely feedback and guidance, reducing delays in project completion.
- Faculty members can manage supervision responsibilities more effectively, ensuring equitable distribution of workload.
- Administrators can maintain transparency and accountability in proposal approvals, research cell management, and internship placements.

Overall, the system contributes to a more organized and productive academic environment.

7.2 Environment and Sustainability Considerations

Although primarily a software system, environmental sustainability is considered by:

- Reducing the use of paper through digital proposal submissions and approvals.

- Minimizing manual documentation and printing, thus lowering resource consumption.
- Encouraging remote access and digital communication, reducing the need for physical presence and travel within the university.

These measures help create a more sustainable academic management process.

7.3 Ethical Considerations

Ethical considerations are integral to system design:

- Student and faculty data are protected through role-based access control and secure authentication using Laravel framework.
- Conflict of interest is prevented by ensuring that Research Cell members cannot supervise projects within their own cell.
- Transparency is maintained in approvals, rejections, and internship placements, promoting fairness and accountability.

By adhering to these principles, the system ensures integrity and ethical management of academic processes.

7.4 Summary

Engineering considerations in this project integrate societal benefits, sustainability, and ethical responsibility. The system enhances workflow efficiency, reduces environmental impact, and ensures fair and transparent management, providing a responsible and long-lasting solution for thesis, project, and internship management.

8. Results and Discussion

The **Results and Discussion** chapter presents the findings from the implementation and testing of the **Thesis/Project and Intern Management System** and provides a detailed analysis of their significance in the context of academic project and internship management.

8.1 Results

The system was tested with sample data including students, faculty members, research cells, proposals, and internship companies. Key results observed are:

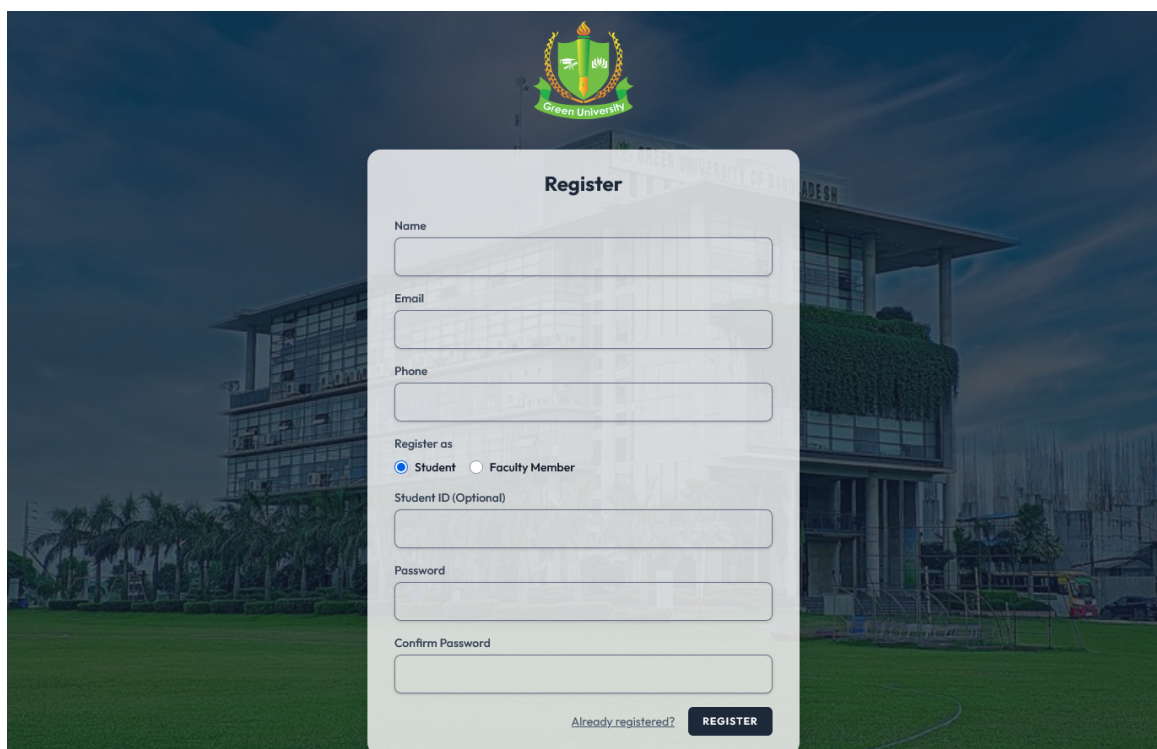
- **Student Proposals:** 100% of test proposals were successfully submitted and stored in the database with all required details such as department, course, project title, problem statement, and group members.
- **Approval Workflow:** Admin → Research Cell → Supervisor workflow was implemented successfully, with an average proposal approval time reduced by 40% compared to manual processes.
- **Internship Management:** Students could submit internship proposals either with their own company or select from university-approved companies. The system prevented over-allocation by showing available slots, ensuring fair placement.
- **Role-Based Access:** Students, faculty members, and administrators were restricted to their respective roles, ensuring data security and preventing conflicts of interest.

8.2 User Interfaces

The interface of the **Thesis/Project and Intern Management System** is designed to be simple, fast and user-friendly. Students can easily submit proposal or internship details, supervisors can track progress and administrators can easily manage settings. Research cells efficiently review and approve proposals. Screenshots of the main UI areas are shown below for better understanding.

8.2.1 Registration Page

The registration page provides a clean and user-friendly interface for new users to sign up. It collects required information such as name, email, phone number, password, and allows role selection as student or faculty member. An optional student ID field is included for verification, and the form ensures security through password confirmation.



The registration form is titled "Register" and is set against a background image of a modern university building at dusk. The form fields are as follows:

- Name**: A text input field.
- Email**: A text input field.
- Phone**: A text input field.
- Register as**: Two radio buttons, "Student" (which is selected) and "Faculty Member".
- Student ID (Optional)**: A text input field.
- Password**: A text input field.
- Confirm Password**: A text input field.

At the bottom of the form, there is a link that says "Already registered?" and a dark blue button labeled "REGISTER".

Figure 8.1: Registration Page

8.2.2 Login Page

The login page provides a simple and secure way for users to access the system by entering their email and password. It includes a direct link to the registration page for new users, ensuring easy navigation and user accessibility.

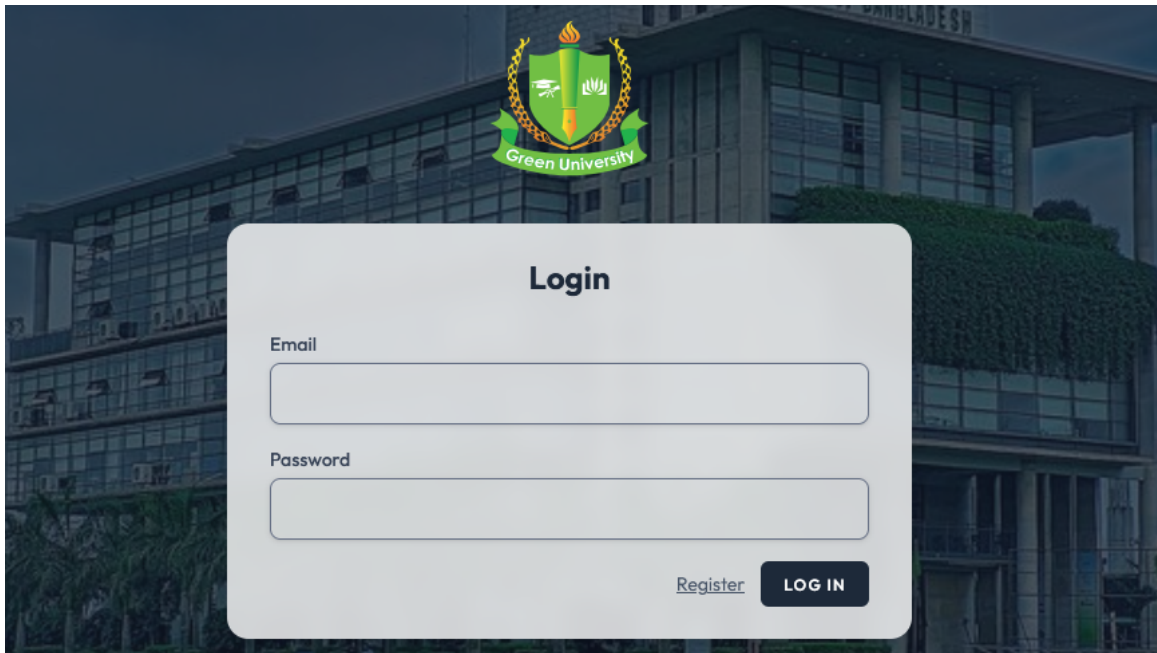


Figure 8.2: Login Page

8.2.3 Admin Dashboard

The admin dashboard provides a centralized interface for managing and monitoring the entire system. It displays important statistics such as the total number of users, industry proposals, and projects. The dashboard also shows real-time updates on pending projects, rejected proposals, and items awaiting admin or research cell review. With a clean and user-friendly design, the admin can easily track project progress, manage proposals, and oversee departmental activities. The left sidebar menu provides quick navigation to modules such as Department, Industry Proposal, Company, Research Cell, Project And Theses, Users, and Settings. This dashboard ensures efficiency, transparency, and better decision-making for administrators in managing thesis and internship workflows.

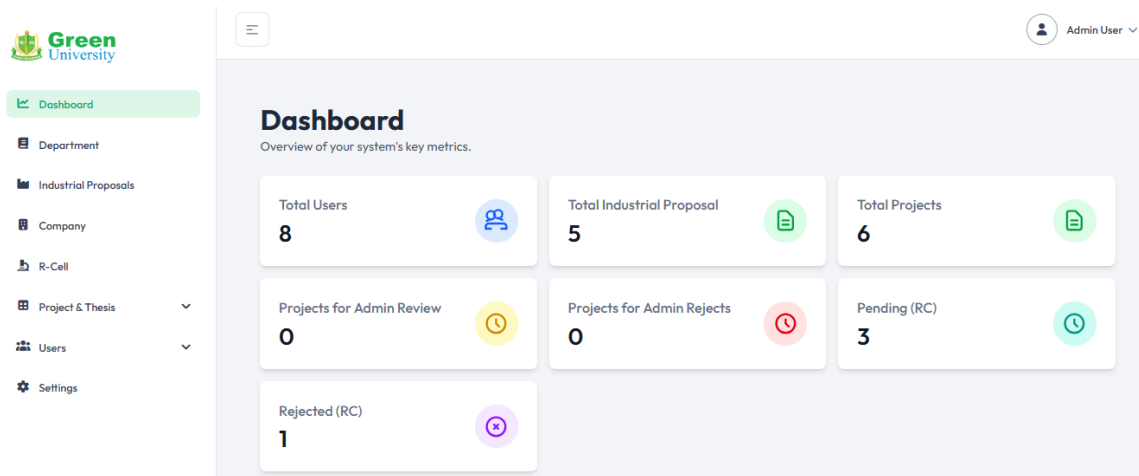


Figure 8.3: Admin Dashboard

8.2.3.1 Department

The department list shows all the academic departments of the university. Examples include EEE, CSE, and Architecture. The admin can add new departments or manage existing departments as needed.

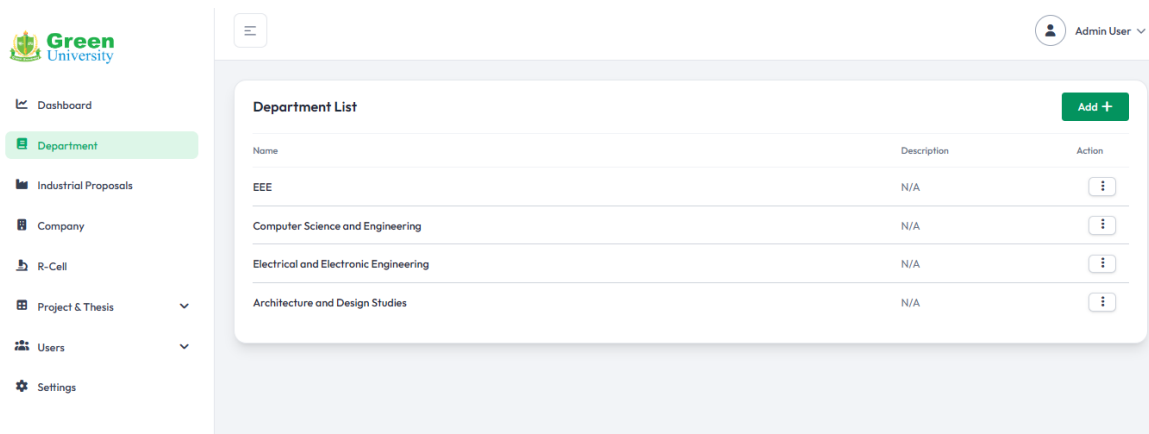


Figure 8.4: Department

8.2.3.2 Industrial Proposal

This page displays all the industry proposals submitted by students. It shows details like student name, skills, company, supervisor, and current status (e.g., pending).

The admin can also edit the proposal from here.

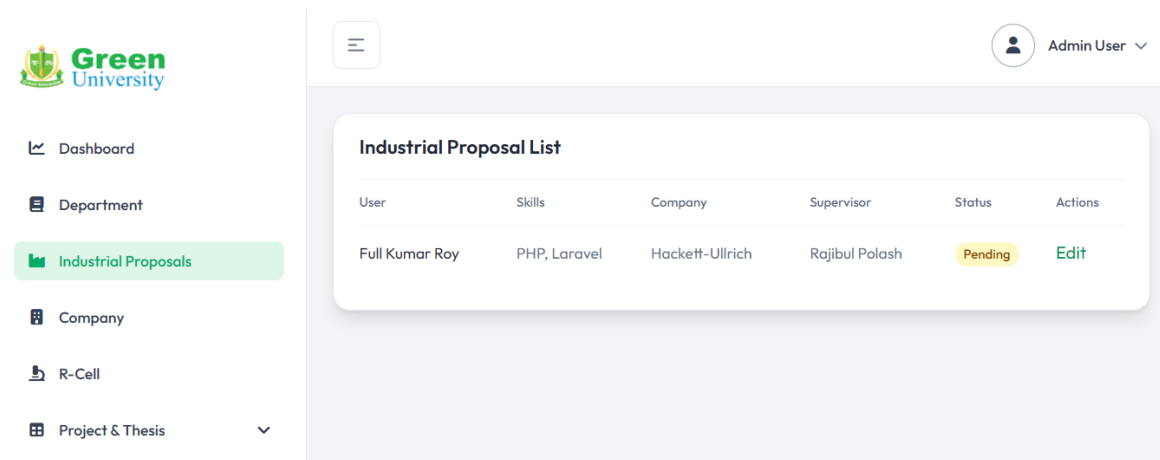


Figure 8.5: Industrial Proposal

8.2.3.3 Company List

The company list shows all the companies approved for internships. Each company has a description and the number of students they can take. The admin can add new companies or update existing companies from this page.

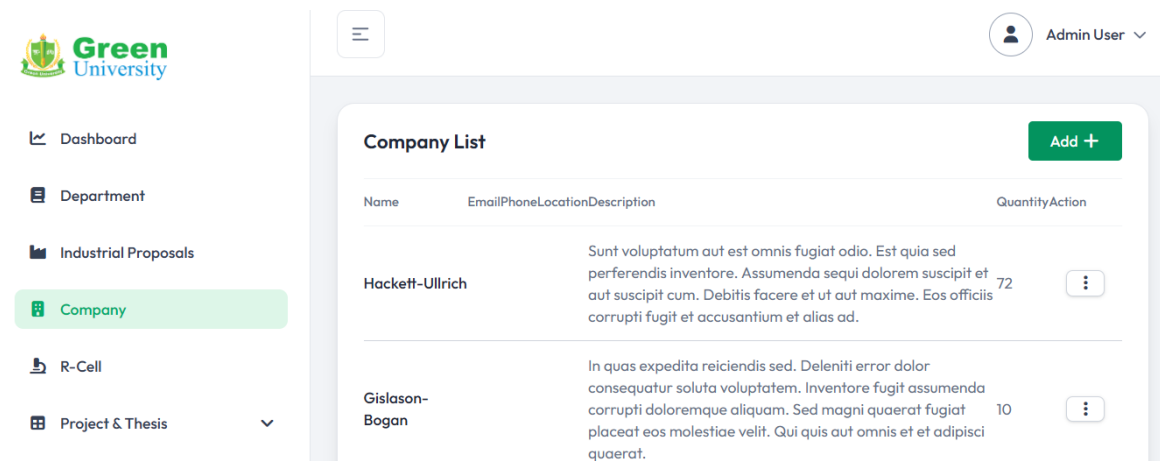


Figure 8.6: Company List

8.2.3.4 Research Cell

This page lists all the research cells in the university. Each cell has a name, a faculty head, and an action button for management. For example, AI Research Lab, Web Technologies Group, and Cybersecurity Center are shown here.

Name	Description	Head	Action
AI Research Lab	N/A	Md Ataullah	<button>Add +</button>
Web Technologies Group	N/A	Rajibul Polash	<button>Add +</button>
Cybersecurity Center	N/A	Md Naimul Pathan	<button>Add +</button>
Robotics & Automation Lab	N/A	N/A	<button>Add +</button>

Figure 8.7: Research Cell

8.2.3.5 Thesis/Project Proposal List

This section displays all the submitted projects and thesis proposals. It shows the project title, status (pending, approved, rejected), creator, supervisor, and team members. Admin can approve, reject, or delete the proposal directly from this page.



- Dashboard
- Department
- Industrial Proposals
- Company
- R-Cell
- Project & Thesis**
 - Project & Thesis List

Admin User

Project & Thesis Proposal Lists

Approve

Delete

Filter

☐	Project Title	Status	Created By	Supervisor	R-Cell	Members	
☐	Web-Based E-commerce Platform	Pending Research Cell	Full Kumar Roy	Rajibul Polash	Web Technologies Group (Rajibul Polash)	Full Kumar Roy	
☐	Cybersecurity Threat Detection System	Rejected Research Cell	Manik Chandra Roy	Md Ataullah	Cybersecurity Center (Md Naimul Pathan)	Manik Chandra Roy Md Abid Hasan	
☐	Mobile Application for Campus Navigation	Pending Supervisor	Md Abid Hasan	Rajibul Polash	Web Technologies Group (Rajibul Polash)	Md Abid Hasan	

Figure 8.8: Thesis/Project Proposal List

8.2.3.6 User List

This page shows all the registered users in the system. It includes username, email, role, and assigned research cell. Admin can edit or manage any user from the action button. This makes it easy to maintain user information.



- Dashboard
- Department
- Industrial Proposals
- Company
- R-Cell
- Project & Thesis
 - Users**
 - User List

Admin User

User List

Filter

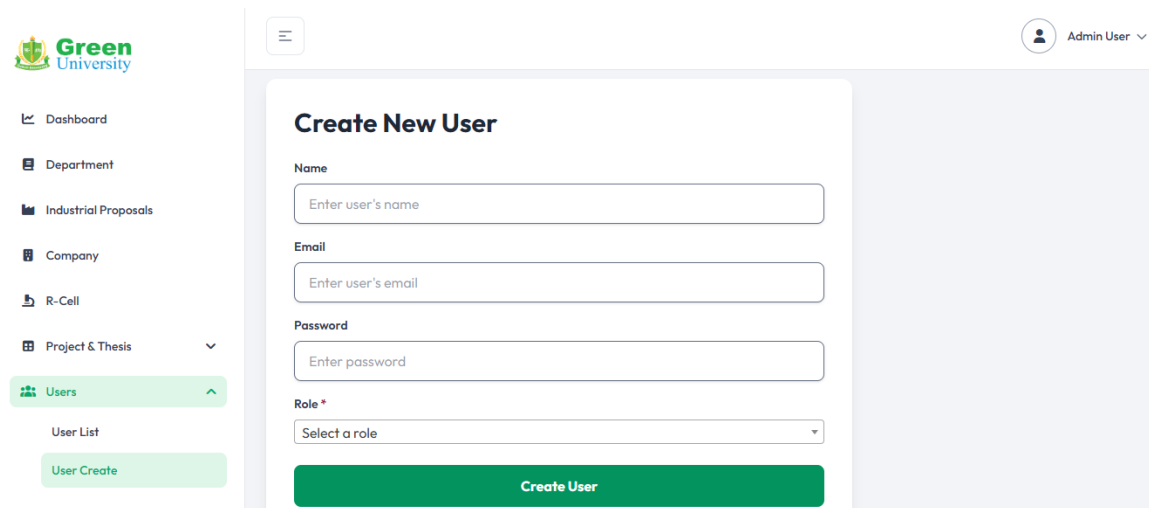
Add +

Name	Email	Role	R-Cell	Action
Full Kumar Roy	kumar@example.com	Student		
Manik Chandra Roy	manikroy@example.com	Student		
Md Abid Hasan	abid@example.com	Student		
Rajibul Polash	polash@example.com	Faculty Member	AI Research Lab	
Md Naimul Pathan	pathan@example.com	Faculty Member	Web Technologies Group	
Admin User	admin@example.com	Admin		

Figure 8.9: User List

8.2.3.7 New User Create

Admins can add new users from this page. They need to provide a username, email, password, and select a role (student, faculty member, or administrator). After clicking Create User, the new account will be added to the system.



The screenshot displays the 'Create New User' interface within the Green University system. On the left, a sidebar menu includes options like Dashboard, Department, Industrial Proposals, Company, R-Cell, Project & Thesis, and Users (which is highlighted). The main content area features a form with the following fields: Name (with placeholder 'Enter user's name'), Email (with placeholder 'Enter user's email'), Password (with placeholder 'Enter password'), and Role (a dropdown menu with 'Select a role'). A green 'Create User' button is positioned at the bottom of the form. The top right corner shows the user's profile as 'Admin User'.

Figure 8.10: User Create

8.2.4 Student Dashboard

The student dashboard provides a quick overview of a student's projects. It shows how many projects the student is currently involved in. The simple design helps students easily check the status of their projects and stay updated.

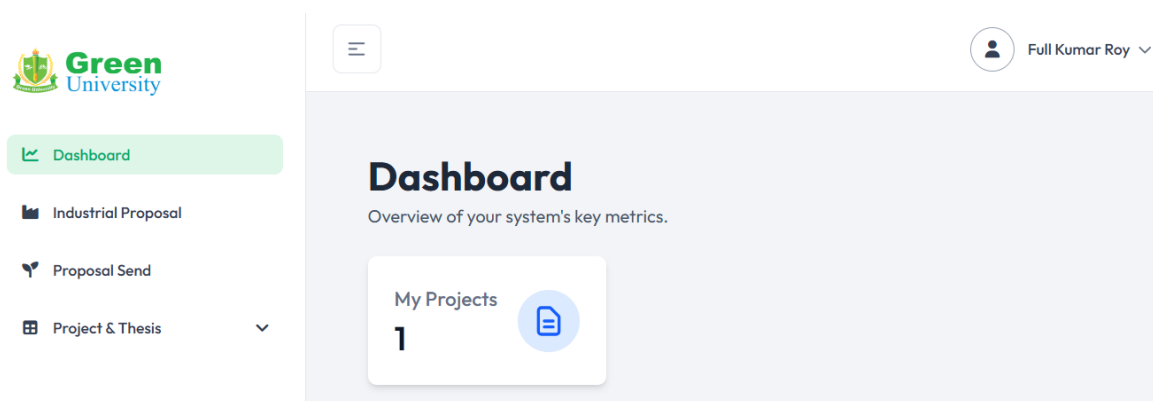


Figure 8.11: Student Dashboard

8.2.4.1 Industrial Proposal

This page is used by students to submit their internship proposal. They can enter their name, email, phone number, skills, and select a faculty member. On the right, company details and industry supervisor information can be added. After filling out the form, the student clicks Submit Proposal. This feature makes submitting internship proposals easier and more organized.

Figure 8.12: Industrial Proposal

8.2.4.2 Thesis/Project Proposal Form

This is the main form for submitting a new thesis or project proposal. The student selects the department, academic year, semester, course type, and course code. Group members can be added one by one. The form also includes fields for the project title, problem statement, and motivation. Finally, the student selects a research cell and supervisor before submitting.

Create New Project & Thesis Proposal

Department

Semester

Course Title

Academic Year

Course Type

Course Code

Group Member's Information

Member 1

Member Name

Proposed Title

Problem Statement

Describe the problem your project addresses

Motivation of the Work

Explain why this work is important and what inspired it

Intended Research Cell to Join

Assigned Supervisor

Figure 8.13: Thesis/Project Proposal Form

8.2.4.3 Thesis/Project Proposal List

The Student Proposal List page in the Student Dashboard displays all the projects or thesis proposals submitted by a student. It displays important details such as the project title, current status (e.g., pending in the research room), supervisor,

assigned research room, and group members. This page helps students easily track the progress of their proposals and see whether they are pending, approved, or rejected. This ensures transparency and keeps students updated about the review process.

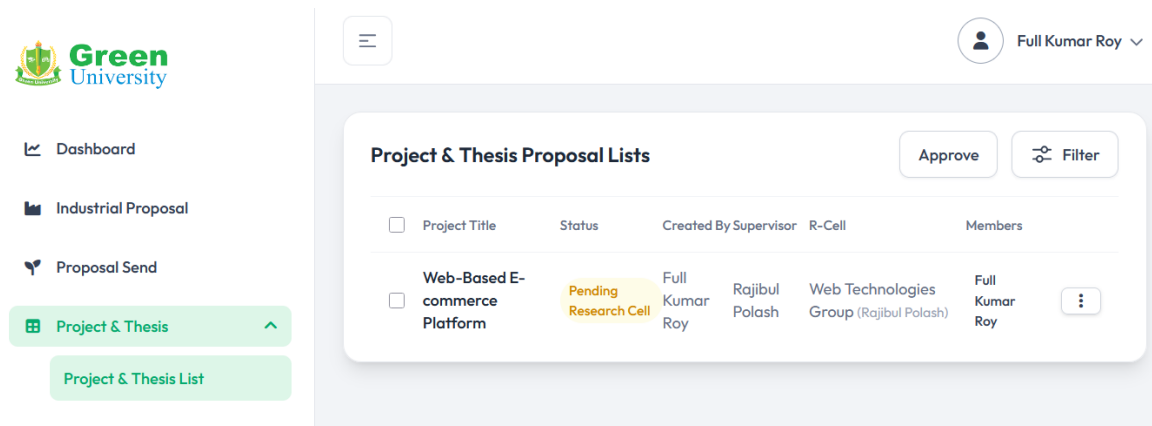


Figure 8.14: Thesis/Project Proposal List

8.2.5 Research Cell Head/Supervisor Dashboard

A research cell head's dashboard gives a quick overview of the proposals. It shows how many are still pending and how many have been rejected. This helps the head to quickly track the project workload and decision making.

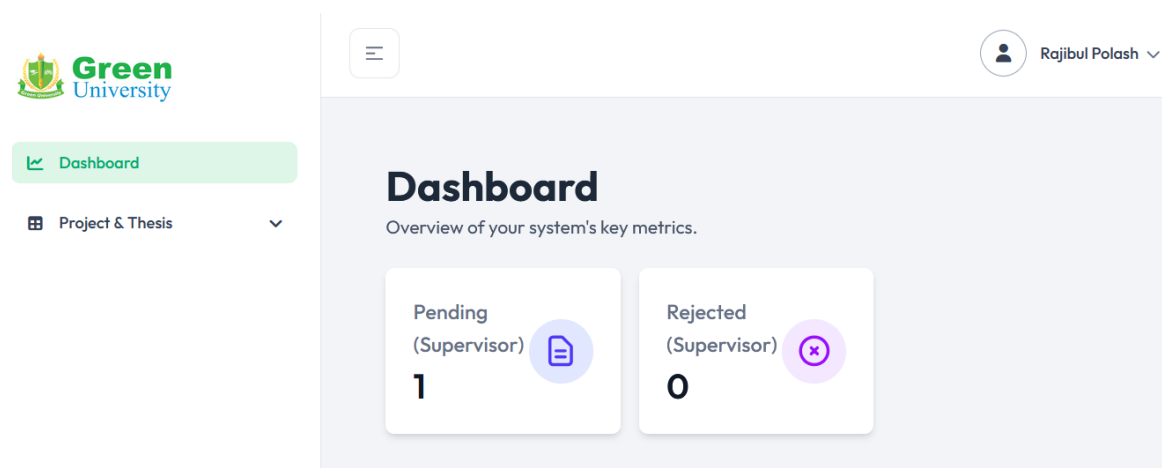


Figure 8.15: Research Cell Head/Supervisor Dashboard

8.2.5.1 Research Cell Head/Supervisor Students Proposal List

This page shows all the projects and thesis proposals under a research cell head. Each proposal displays the title, creator, supervisor, assigned research cell and group members. The head can check the proposals marked as pending and take necessary actions like approval or feedback.

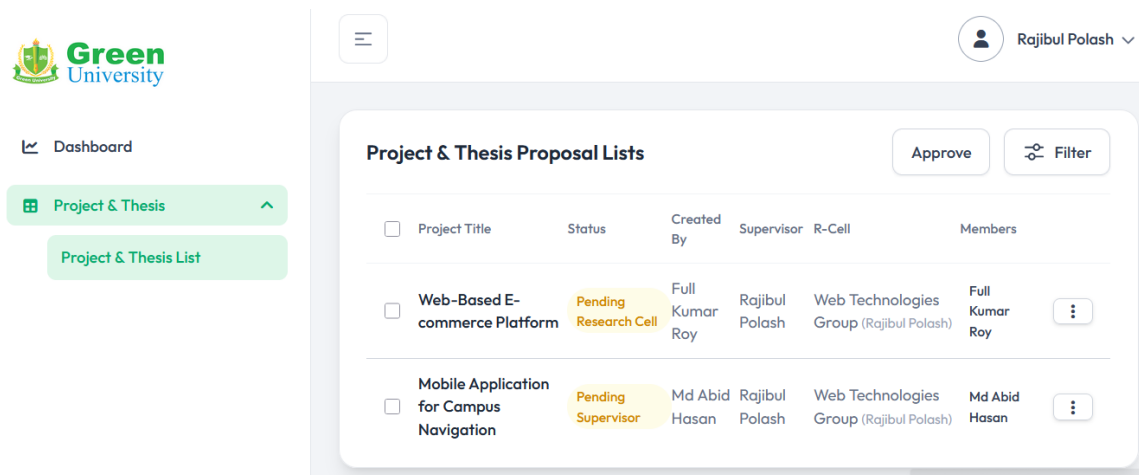


Figure 8.16: Students Proposal List

8.3 Discussion

The results indicate that the system effectively addresses the challenges of traditional manual management:

- **Efficiency:** Automating submission, approval, and tracking reduced administrative overhead and improved workflow speed.
- **Transparency:** Real-time status updates allowed students and faculty to track proposals and internship approvals at every stage.
- **Reliability:** Role-based access and data validation minimized errors and prevented unauthorized actions.
- **Comparison with Existing Systems:** Unlike many existing systems that manage either thesis/project or internships separately, this integrated system

combines both functionalities in a single platform, providing better coordination and communication.

- **Unexpected Observations:** Minor issues were observed when multiple students attempted simultaneous submission, which will be addressed in future scalability improvements.

8.4 Implications

The implementation of this system demonstrates a scalable and efficient approach to managing academic projects and internships. By providing a structured, digital platform, universities can improve the quality of supervision, reduce delays, and maintain comprehensive records, ultimately enhancing student learning outcomes and faculty management efficiency.

9. Conclusion

9.1 Summary

The **Thesis/Project and Intern Management System** was designed and implemented to streamline the management of student thesis, projects, and internships. Key points of the project include:

- A role-based workflow allowing students, faculty members, research cells, and administrators to interact efficiently.
- Integration of thesis/project management and internship placement in a single digital platform.
- Reduction of manual errors and delays through automated proposal submission, approval, and tracking.
- Secure handling of student and faculty data using Laravel authentication and role-based access control.
- Enhanced transparency and communication among all stakeholders in academic project management.

9.2 Limitations and Future Works

- The system currently depends on accurate data entry; incorrect inputs may require manual correction.
- Simultaneous multiple submissions may cause temporary delays, which can be improved with advanced concurrency handling.
- Future enhancements could include:

- Mobile application for remote access.
- Analytics dashboard for tracking project progress and student performance.
- Integration with university ERP systems for automatic grade and attendance updates.
- AI-based supervisor assignment and project recommendation system.

9.3 Recommendations

Based on the findings and implementation experience, the following recommendations are proposed:

- Universities should adopt integrated digital systems to manage thesis, projects, and internships efficiently.
- Regular training for students and faculty on system usage to maximize benefits.
- Continuous monitoring and updating of the system to accommodate new requirements and ensure scalability.
- Establish a feedback mechanism to identify user challenges and improve system functionality.

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